



TRACKSHIFT 2026 · AI MOTORSPORT INTELLIGENCE · TYRE DEGRADATION INTELLIGENCE

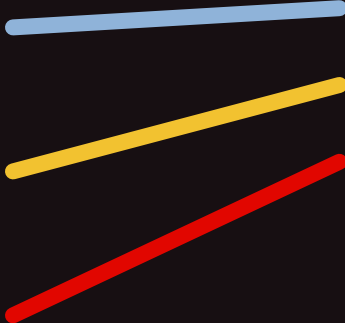
ClearStint

Practice tyre data does not match the race. ClearStint learns the gap.

We fit degradation curves from practice, learn a practice-to-race correction each weekend, and flag when there is not enough data to trust a curve.

Working prototype · 1,544 practice laps and 3,177 race laps after quality filtering · 6 Grands Prix of 2026 · public data only · nothing tuned on the race

Team CleanStint · Samuel Christ & Kaushal R · Idea submission, 5 September 2026





How wrong is Friday?

7x

Hungary 2026, medium tyre. A straight line through Friday's long-run laps says it degrades at +0.28 s per lap. The race showed +0.04 s.

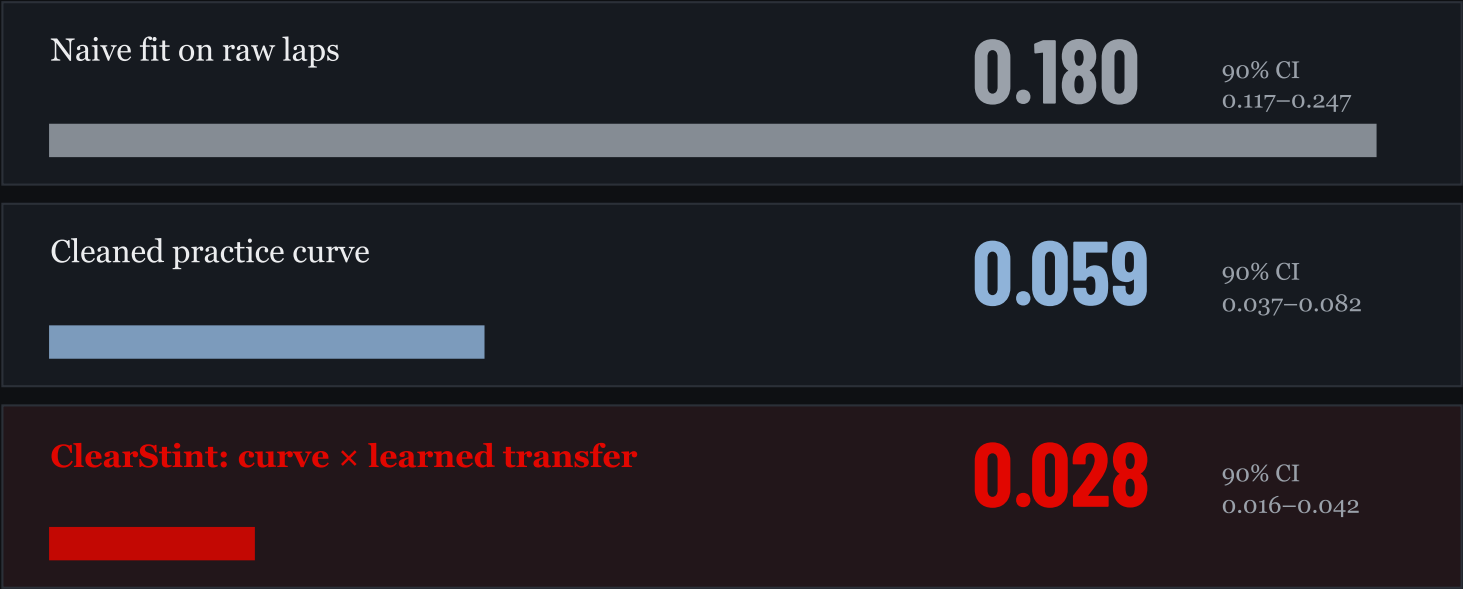
Because a lap time is fuel burn + track evolution + traffic + driver + tyre.
We only want the tyre part.

Even after cleaning, the practice curve is still steeper than the race.

Practice and race stress the tyre differently in ways a lap-count model cannot see. On the medium, race degradation ran at roughly 0.4 of the cleaned practice value on every European weekend we tested. We have not found this ratio published anywhere, so we estimate it and update it after each race.

Tested on 6 Grands Prix, leave-one-weekend-out

Fit on practice only. Transfer factors learned from the other weekends. Scored against the race with the same estimator. Mean absolute error in degradation, s/lap per lap of tyre age, with a 90% bootstrap interval.



Probability that the transfer step lowers error (bootstrap over weekends): 99%. Against a single global factor for all compounds, ClearStint wins 9 of 12.

By compound

	n	Clean	ClearStint	Factor
Soft	5	0.048	0.048	none
Medium	5	0.089	0.014	×0.43
Hard	2	0.010	0.010	none

A factor is applied only where a majority of other weekends agree within ±50%. Today that is the medium: its error falls from 0.089 to 0.014 (90% CI on the gain 0.036 to 0.114). Soft ratios disagree between circuits and hard has two weekends, so their cleaned curves stand unchanged.

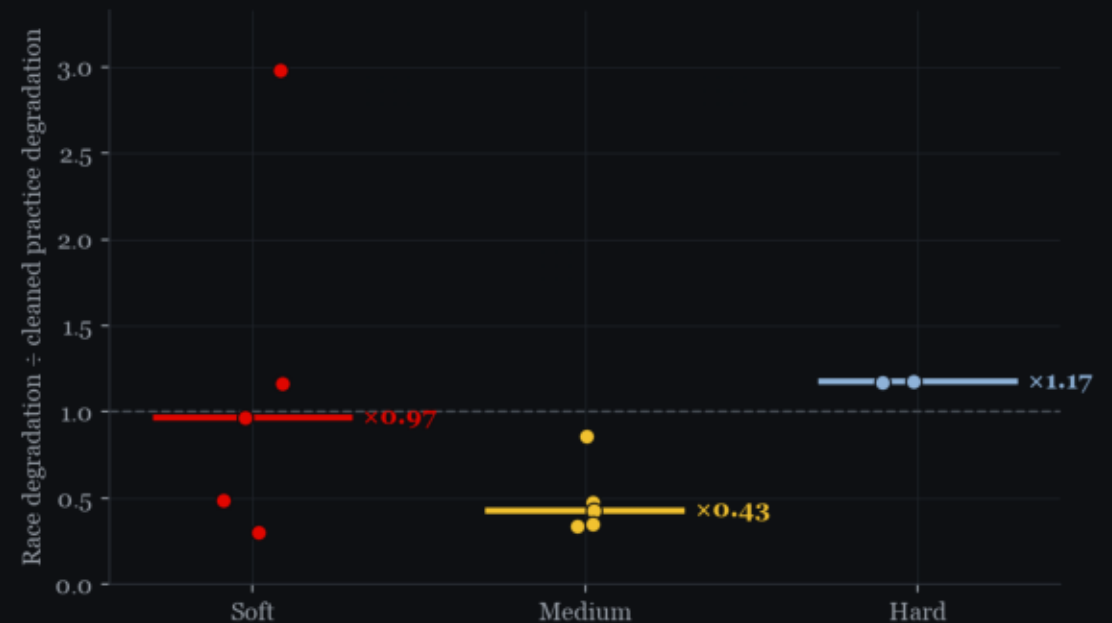
Withheld and reported: 3 compound-weekends with no positive cleaned slope (Australia hard, Australia medium, Japan hard; season-opening rounds) and 2 with under 30 practice laps. Japan soft had no usable race stint, so 17 of 18 are counted.

The medium is where practice and race disagree most

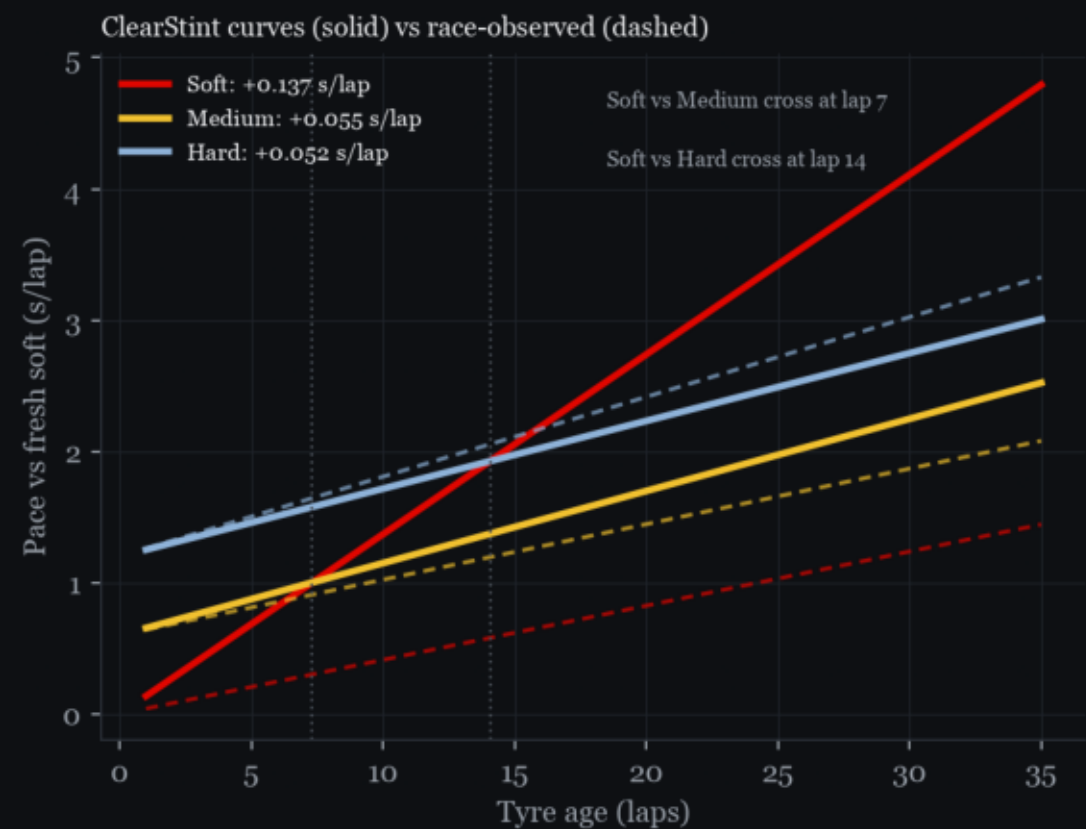
Race degradation divided by cleaned practice degradation, one dot per weekend. The medium sits at $\times 0.35$ to $\times 0.47$ on every European weekend (Japan $\times 0.86$). Soft scatters from $\times 0.30$ to $\times 3.0$; Hard has two points near $\times 1.2$. So the model corrects the medium and leaves the others alone, and it says so.

Why the gap exists

We measured tyre energy per lap from the public 3.7 Hz position and speed traces. At Austria, the one circuit checked so far, practice and race laps carry the same energy (+2%), so the gap does not look like drivers simply pushing harder on Friday. The medium is the compound teams manage on Sunday. A pit wall reading Friday's medium curve at face value plans stints that are far too short. The next slide shows what that costs in a real stint plan.



Saturday night, Hungary 2026: the decision



Curve used	Best plan	Stint laps	Cost vs best
Naive fit	M-H-H	20/24/26	+74 s
Cleaned curve	S-S-H	20/20/30	+35 s
ClearStint	S-M-M	14/28/28	+26 s
Race-observed	S-S-M	28/28/14	0 (best)

What the field did: 2-stop for 12 of 19 classified drivers.
Most common sequences: M-H-H (6), M-H-S (4), S-M-S (2).
Median stint length: Hard 22, Medium 20, Soft 14 laps.

Pit loss 21 s; compound pace steps assumed 0.6 s (soft < medium < hard); linear degradation. 'Cost vs best' = extra race time if you follow that curve's plan and the race behaves as observed.

Follow the naive curve and the race costs you +74 s against the best plan; the cleaned curve +35 s; ClearStint +26 s. Compound offsets are an assumption today and come from qualifying on Challenge Day. The race-observed plan is a hindsight optimum under the observed curves; no team ran it. Crossovers under ClearStint: Soft vs Medium at lap 7, Soft vs Hard at lap 14.

How it works

1 Ingest

FastF1 + OpenF1: laps, stints, compound, tyre age, sectors, weather, 3.7 Hz speed and position, gap to car ahead

2 Quality-check

Per-lap telemetry health. The Hungary race position feed has ≈ 26 distinct points per lap; we detect that and fall back to timing-only features

3 Strip confounders

Fuel prior (2026: ≈ 1.1 kg/lap) $\cdot$ track evolution from every driver's push laps $\cdot$ traffic laps dropped via gap data $\cdot$ per-stint effects absorb driver and fuel load

4 Clean curve

Degradation per compound from practice alone. Fixed-effects fit today; hierarchical Bayesian with credible bands on Challenge Day

5 Transfer

Race $\div$ practice ratio per compound from the other weekends (leave-one-weekend-out), applied only where they agree

6 Validate

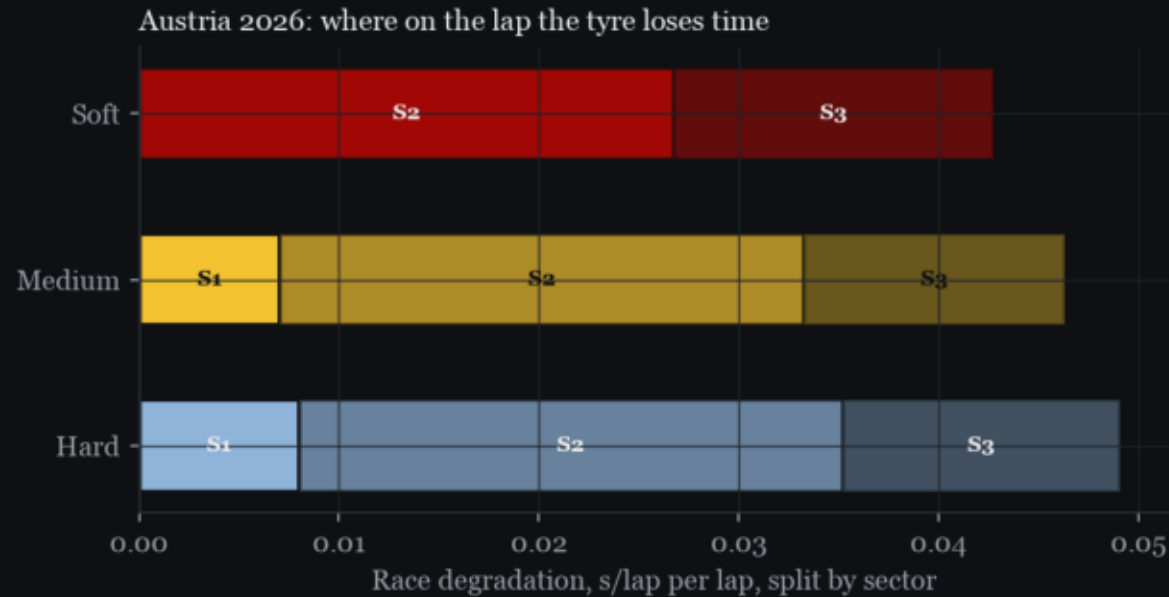
After every race: predicted vs observed, per compound and per sector. The scorecard updates the factors

$$\text{lap_time} = \text{stint_effect} + \text{deg}[\text{compound}] \cdot \text{tyre_age} + \beta \cdot \text{fuel} + \text{evo}(\text{session_time}) + \varepsilon$$

$$\text{race_deg} = k[\text{compound}] \cdot \text{deg}[\text{compound}]$$

Stack: Python 3.12, pandas, NumPy, FastF1 3.8, OpenF1; PyMC for the Bayesian model; scikit-learn; Streamlit dashboard. Everything here is disclosed pre-work; Challenge Day adds the Bayesian fit, a non-linear cliff term, track temperature as a covariate, sprint weekends, and the dashboard.

What the pit wall gets



Sector times split the degradation by part of the lap. A front-limited or traction-limited tyre shows up here on Friday.

What the dashboard shows

- Race-calibrated degradation curve per compound, with a band and the transfer factor shown, or 'no reliable curve' when the data does not support one.
- Crossover laps and a 1-stop vs 2-stop comparison, as on the previous slide, for the coming race.
- The list of laps excluded and why: traffic, flag, aborted, degraded telemetry.
- Sector-level degradation map: where on the lap the tyre is giving up time.

Sunday-night scorecard

Predicted vs observed per compound. It updates the transfer factors, so the factors improve as the season goes on.



Off the track: the battery in an e-rickshaw

An e-rickshaw's lead-acid battery pack costs ₹40,000 to ₹60,000 and is replaced every 12 to 18 months; a lithium pack ₹60,000 to ₹1,20,000 every 3 to 5 years. That is the single largest running cost for the more than a million e-rickshaws on Indian roads, and operators replace on a calendar, not on measured health. Replacing six months early wastes about ₹20,000 per vehicle; replacing late strands a driver mid-shift.

	Confounders to strip	'Friday' vs 'Sunday'	Curve wanted
F1 tyre	Fuel, track evolution, traffic, driver	Practice long run vs managed race stint	Pace loss per lap
E-rickshaw battery	Temperature, load, charging habit, route	Lab cycle test vs real daily duty	Capacity loss per cycle

Same model: remove the confounders, learn the lab-to-road correction, and flag low-confidence cases. First test: the public Severson et al. (2019) dataset (124 cells, charging-policy confounders; metric: predicted vs measured capacity). Then a pilot with an e-rickshaw fleet operator or leasing lender in Punjab, where Plaksha sits. This would let drivers replace batteries on measured health instead of a fixed schedule.



Summary



Built on public data only

3.7 Hz telemetry, no fuel readings, one race feed broken at source. Where data is missing we estimate it and flag it.



Measurable

Across 6 Grands Prix, leave-one-weekend-out, degradation error falls from 0.180 to 0.028 s/lap. On the Hungary replay, the naive plan costs +74 s; ours +26 s.



Withholds when unsure

5 of 17 compound-weekends withheld and reported. Transfer factors applied only where weekends agree. A strategist or a lender needs to know when the number cannot be trusted.



Feasible, and India-bound

The pipeline, quality checks and validation harness all run today. Challenge Day is the Bayesian model and the dashboard. The same loop then goes to e-rickshaw batteries, with a public dataset already chosen.